

Criticality Thresholds in One-Dimensional Multiplying Media with n -Bonacci Aperiodic Modulation

Spectral Gap Control of k_{eff} in Substitution-Sequence Diffusion Operators

Pablo Nogueira Grossi¹

¹G6 LLC, Newark, New Jersey, USA, pablogrossi@hotmail.com, ORCID: 0009-0000-6496-2186

2026

Abstract

We study the one-group neutron diffusion equation on a uniform one-dimensional slab whose material coefficients—diffusion constant, removal cross-section, and fission production rate—vary site-by-site according to the n -bonacci substitution sequence for $n = 2, 3, 4, 5$. The criticality condition $k_{\text{eff}} = 1$ defines a critical fission strength $\lambda_c(n)$, which we compute by solving the associated generalized eigenvalue problem via finite differences. We find that $\lambda_c(n)$ is governed not by the n -bonacci constant ρ_n alone, but by the *spectral gap* of the substitution transfer matrix, $\Delta_n = \rho_n - |\rho_n^{(2)}|$, where $\rho_n^{(2)}$ is the subdominant root of the characteristic polynomial $x^n = x^{n-1} + \dots + 1$. Numerically, $\lambda_c(n) \approx 0.958 \Delta_n + 0.107$ with correlation $r = 0.989$ across $n = 2, \dots, 5$. For $n \geq 4$, $\lambda_c(n)$ converges to $7/6$ exactly; the Tribonacci case $n = 3$ saturates at the distinct value $\lambda_c(3) \approx 37/32$, while $\lambda_c(2) \approx 1.064$ for Fibonacci. These results extend and complement a companion study of nonlinear (DNLS) dynamics on Fibonacci and Tribonacci chains [5], and provide a concrete spectral-theoretic framework for aperiodic multiplying media.

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1 Introduction

The interplay between aperiodic order and spectral properties of differential operators has been a central theme in mathematical physics since the discovery of quasicrystals [10] and the subsequent rigorous treatment of almost-periodic Schrödinger operators [1, 3]. The Fibonacci chain—generated by the substitution $A \mapsto AB$, $B \mapsto A$ —is the canonical one-dimensional model: its tight-binding Hamiltonian has a Cantor-set spectrum of measure zero, multifractal eigenstates, and spectral gaps labelled by the gap-labelling theorem [1]. The Tribonacci chain, generated by $A \mapsto AB$, $B \mapsto AC$, $C \mapsto A$, extends this framework to three letter alphabets and introduces richer hierarchical structure governed by the tribonacci constant $\rho_3 \approx 1.839$ [7].

Transport theory in heterogeneous media provides a physically motivated arena for the same mathematical structures. The neutron diffusion equation in a multiplying medium,

$$-\nabla \cdot (D(\mathbf{r}) \nabla \phi) + \Sigma_r(\mathbf{r}) \phi = \frac{1}{k_{\text{eff}}} \nu \Sigma_f(\mathbf{r}) \phi, \quad (1)$$

defines k_{eff} as the dominant eigenvalue of the ratio of fission production to neutron loss. When the material coefficients D , Σ_r , $\nu \Sigma_f$ are spatially uniform, k_{eff} is determined by simple geometric buckling. When they vary periodically, Bloch theory applies and band-gap phenomena are well understood [11]. When they vary *aperiodically*—as in a substitution-sequence medium—neither framework is directly applicable, and the spectral theory of the associated operator must be studied on its own terms.

1.1 Motivation and relation to prior work

This paper is motivated by two converging lines of inquiry.

The first is a companion study [5] in which the discrete nonlinear Schrödinger (DNLS) equation is evolved on Fibonacci and Tribonacci hopping chains. There, the mid-gap eigenstates are multifractal—a hallmark of critical (neither localized nor extended) behavior—and the inverse participation ratio (IPR) responds differentially to nonlinearity: Fibonacci and Tribonacci chains exhibit distinct robustness thresholds as the coupling λ increases. The present work asks the same structural question in a transport setting: *how does the substitution rule governing spatial heterogeneity control the criticality threshold of a multiplying medium?*

The second motivation is the gap-labelling program for one-dimensional operators with substitution-sequence potentials. The spectral gap between the dominant and subdominant eigenvalues of the transfer matrix—what we call $\Delta_n = \rho_n - |\rho_n^{(2)}|$ —governs the rate at which finite-generation approximants converge to the thermodynamic limit. We find, numerically, that Δ_n also directly controls $\lambda_c(n)$, suggesting a precise connection between the transfer-matrix spectral gap and the physical criticality threshold. This connection is the central result of the paper.

1.2 Main results

We establish the following, numerically for $n = 2, 3, 4, 5$ and analytically for the linear relationship in the large- N limit:

1. **Spectral gap governs criticality.** The critical fission strength $\lambda_c(n)$ satisfies

$$\lambda_c(n) \approx \alpha \Delta_n + \beta, \quad \alpha \approx 0.958, \quad \beta \approx 0.107, \quad (2)$$

with correlation $r = 0.989$ across the n -bonacci family $n = 2, \dots, 5$.

2. **Saturation and exact values.** For $n \geq 4$, $\lambda_c(n) \rightarrow 7/6$ exactly (verified to machine precision for $n = 5$, converging from below for $n = 4$). The Tribonacci case $n = 3$ saturates at the distinct value $\lambda_c(3) \approx 37/32 = 1.15625$, stable across generations $g = 7-9$. The Fibonacci case $n = 2$ gives $\lambda_c(2) \approx 1.064$, the lowest threshold, reflecting the minimal absorber content of the two-symbol alphabet. The existence of exact rational limits for $n \geq 3$ suggests an analytic explanation via the transfer-matrix fixed-point structure.
3. **Fibonacci as outlier.** The two-letter Fibonacci substitution ($n = 2$) occupies a qualitatively distinct regime from the $n \geq 3$ family. Its spectral gap $\Delta_2 = \rho_2 - |\rho_2^{(2)}| = 1$ is the only integer value among $n = 2, \dots, 5$, and its critical threshold $\lambda_c(2) \approx 1.064$ is the lowest, reflecting the minimal absorber content of the two-symbol alphabet.
4. **Generation convergence.** k_{eff} converges to its thermodynamic-limit value with generation g at a rate consistent with the spectral gap of the substitution matrix. Tribonacci ($n = 3$) converges more slowly than Fibonacci ($n = 2$), consistent with its smaller Δ_n .

1.3 Organization

Section 2 introduces the n -bonacci substitution sequences, the one-group diffusion model, and the finite-difference discretization. Section ?? reviews the transfer-matrix spectral theory that motivates the λ_c - Δ_n relationship. Section ?? presents the numerical results: the $\lambda_c(n)$ table, the spectral gap correlation, the saturation phenomenon, and generation convergence. Section ?? examines the fundamental flux modes and their hierarchical spatial structure. Section ?? discusses connections to the DNLS companion paper, the gap-labelling theorem, and open questions.

2 Model

2.1 The n -bonacci substitution sequences

For each integer $n \geq 2$, the n -bonacci substitution acts on an alphabet $\mathcal{A}_n = \{A_0, A_1, \dots, A_{n-1}\}$ by the rules

$$A_0 \mapsto A_0 A_1, \quad A_k \mapsto A_0 A_{k+1} \quad (1 \leq k \leq n-2), \quad A_{n-1} \mapsto A_0. \quad (3)$$

Beginning from the seed word A_0 , iterating the substitution g times produces a word $\mathbf{w}^{(n,g)}$ of length $|\mathbf{w}^{(n,g)}|$ that grows as ρ_n^g where ρ_n is the unique real root greater than one of the characteristic polynomial

$$p_n(x) = x^n - x^{n-1} - x^{n-2} - \dots - x - 1. \quad (4)$$

Definition 2.1 (*n*-bonacci constant). *The n-bonacci constant ρ_n is the unique positive real root of $p_n(x) = 0$ with $\rho_n > 1$.*

The cases $n = 2$ and $n = 3$ recover the familiar Fibonacci ($\rho_2 = (1 + \sqrt{5})/2 \approx 1.618$) and Tribonacci ($\rho_3 \approx 1.839$) substitutions respectively. Table 1 collects the constants and chain lengths used in this paper.

Table 1: *n*-bonacci constants, subdominant root moduli, spectral gaps, and chain lengths at generation g used in numerical experiments.

n	ρ_n	$ \rho_n^{(2)} $	Δ_n	g	$N = \mathbf{w}^{(n,g)} $
2	1.61803	0.61803	1.00000	10	144
3	1.83929	0.73735	1.10193	8	149
4	1.92756	0.81828	1.10929	8	208
5	1.96595	0.87105	1.09490	8	236

2.2 The one-group diffusion model

We consider a one-dimensional slab of length $L = N h$ (where h is the mesh spacing, set to $h = 1$ throughout) occupied by N sites. Each site $i \in \{1, \dots, N\}$ carries material parameters determined by the symbol $w_i \in \mathcal{A}_n$ at position i in the substitution word $\mathbf{w}^{(n,g)}$:

$$(D_i, \Sigma_r^{(i)}, \nu \Sigma_f^{(i)}) = \begin{cases} (1.0, 0.5, \lambda) & \text{if } w_i = A_0 \text{ (fissile),} \\ (1.0, 2.0, 0) & \text{if } w_i \in \{A_1, \dots, A_{n-1}\} \text{ (absorber).} \end{cases} \quad (5)$$

The parameter $\lambda \geq 0$ is the fission production strength and is the sole free parameter of the model. All other values are dimensionless and fixed; they are chosen so that (i) fissile and absorber sites differ sharply in both removal cross-section ($\Sigma_r^{(0)} = 0.5$ vs. $\Sigma_r^{(k)} = 2.0$) and fission content, creating genuine spectral competition between production and loss, and (ii) the system crosses criticality ($k_{\text{eff}} = 1$) at a value of λ of order unity, making finite-size effects observable across reasonable generation numbers.

Remark 2.2. *The model can be interpreted as a one-speed diffusion approximation in a heterogeneous multiplying medium with aperiodically modulated coefficients, in the spirit of quasicrystal-inspired transport problems. The substitution word encodes a deterministic aperiodic disorder rather than a random or periodic one; this is the key distinction from classical homogenization theory.*

The one-group diffusion equation (1) on this domain, with vacuum boundary conditions $\phi(0) = \phi(L) = 0$, becomes the eigenvalue problem

$$L \phi = \frac{1}{k_{\text{eff}}} F \phi, \quad (6)$$

where $\phi = (\phi_1, \dots, \phi_N)^T$ is the discretized flux vector, L is the loss (diffusion plus removal) matrix, and F is the fission matrix defined below.

2.3 Finite-difference discretization

We discretize equation (1) by standard cell-centered finite differences on the uniform mesh $x_i = (i - 1/2)h$, $i = 1, \dots, N$. At interior cell interfaces $x_{i+1/2}$, the diffusion coefficient is approximated by the harmonic mean

$$D_{i+1/2} = \frac{2 D_i D_{i+1}}{D_i + D_{i+1}}, \quad (7)$$

which is the standard choice for discontinuous diffusion coefficients as it ensures continuity of the neutron current $J = -D d\phi/dx$ across material interfaces [4]. At the boundaries, vacuum conditions are implemented by setting $D_{1/2} = D_1$ and $D_{N+1/2} = D_N$.

The loss matrix $L \in \mathbb{R}^{N \times N}$ is symmetric tridiagonal:

$$L_{ii} = D_{i-1/2} + D_{i+1/2} + \Sigma_r^{(i)}, \quad (8)$$

$$L_{i,i\pm1} = -D_{i\pm1/2}. \quad (9)$$

The fission matrix $F = \text{diag}(\nu\Sigma_f^{(1)}, \dots, \nu\Sigma_f^{(N)})$ is diagonal and non-negative. Since the only non-zero entries of F occur at fissile sites ($w_i = A_0$), F is in general rank-deficient, with $\text{rank}(F)$ equal to the number of A_0 symbols in $\mathbf{w}^{(n,g)}$.

The eigenvalue problem (6) is equivalent to

$$F \phi = k_{\text{eff}} L \phi, \quad (10)$$

which is a generalized eigenvalue problem with L symmetric positive definite and F symmetric positive semidefinite. By the Perron–Frobenius theorem applied to $L^{-1}F$, the dominant eigenvalue k_{eff} is real, positive, and simple, with a strictly positive eigenvector (the fundamental flux mode) [12].

We compute k_{eff} and the fundamental mode using `scipy.linalg.eig` applied to the dense matrices, which is feasible for $N \leq 300$. For reproducibility, all code is available at [6].

2.4 The critical fission strength $\lambda_c(n)$

For fixed n and generation g , $k_{\text{eff}}(\lambda)$ is a strictly increasing, continuous function of λ (since increasing the fission production at fissile sites monotonically increases the dominant eigenvalue of $L^{-1}F$). We define the *critical fission strength* as the unique root of

$$k_{\text{eff}}(\lambda) = 1, \quad (11)$$

which is computed by Brent’s method [2] bracketed on $[\lambda_{\min}, \lambda_{\max}] = [0.3, 6.0]$.

The quantity $\lambda_c(n, g)$ depends on g through the finite-size geometry of $\mathbf{w}^{(n,g)}$, and converges as $g \rightarrow \infty$ to a thermodynamic-limit value $\lambda_c(n) := \lim_{g \rightarrow \infty} \lambda_c(n, g)$. We study this convergence in Section ?? and use the largest tractable generation as a proxy for the thermodynamic limit.

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